

Question_5:

Code:

```
1 // An undirected graph receives edges through queries. Add an edge only if every
2 // simple cycle in the graph has XOR value 1. Print YES if it can be added, NO other
3 #include <iostream>
4 using namespace std;
5
6 #define MAXN 300005
7
8 int parentNode[MAXN];
9 int weightXor[MAXN];
10
11 void initDSU(int n)
12 {
13     for (int i = 1; i <= n; i++)
14     {
15         parentNode[i] = i;
16         weightXor[i] = 0;
17     }
18 }
19 int findRoot(int i, int *xorSum)
20 {
21     if (parentNode[i] == i)
22     {
23         *xorSum = 0;
24         return i;
25     }
26
27     int parentXor = 0;
28     int root = findRoot(parentNode[i], &parentXor);
29     *xorSum = weightXor[i] ^ parentXor;
30
31     return root;
32 }
33
34 int dfs1(int np, int lst)
35 {
36     return np;
37 }
38 int main()
```

```
39 {
40     int n, q;
41     cin >> n >> q;
42
43     initDSU(n);
44     while (q--)
45     {
46         int u, v, x;
47         cin >> u >> v >> x;
48
49         int xorU = 0, xorV = 0;
50         int rootU = findRoot(u, &xorU);
51         int rootV = findRoot(v, &xorV);
52
53         if (rootU != rootV)
54         {
55             parentNode[rootU] = rootV;
56             weightXor[rootU] = xorU ^ xorV ^ x;
57             cout << "YES\n";
58         }
59         else
60         {
61             int cycleXor = xorU ^ xorV ^ x;
62             cout << (cycleXor == 1 ? "YES" : "NO") << "\n";
63         }
64     }
65     return 0;
66 }
```

Output:

Test Case_1:

```
9 12
6 1 0
YES
1 3 1
YES
3 6 0
YES
6 2 0
YES
6 4 1
YES
3 4 1
YES
2 4 0
YES
2 5 0
YES
4 5 0
YES
7 8 1
YES
8 9 1
YES
9 7 0
NO
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
9 6
1 3 1
YES
3 6 0
YES
6 4 1
YES
2 4 0
YES
4 5 0
YES
7 8 1
YES
PS C:\Users\Prabin Yadav\
```